

Bayesian Data Analysis

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I will introduce the Bayesian approach to data analysis.

We start from scratch.

I will focus on concepts and proceed by way of examples.

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The less statistics you know, the less you need to "unlearn" to break into Bayesian data analysis.

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There will be four parts:

1. Bayesian model (3 sessions)
2. Bayesian inference (3 sessions)
3. Bayesian hierarchical model (.5 session)
4. Bayesian model checking (.5 session)

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Part 1: Bayesian Model

Session 1: Mostly motivation
Session 2: Mostly definitions
Session 3: Mostly computation and examples

Part 2: Bayesian Inference

Session 4: Posterior as estimate of θ and summaries
Session 5: Simulation based inference about θ and $g(\theta)$
Session 6: Simulation based prediction of $\tilde{y}$ and $h(\tilde{y})$

Parts 3, 4: Hierarchical Model and Model Checking

Session 7: Bayesian hierarchical model and model checking

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Resources available are:

- Class notes
- Slides used in class
- R scripts with examples
- JAGS documentation
- References

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